

Final Project Guidelines

Monte Carlo Methods

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Philosophy

“The only way to learn mathematics is to do mathematics.”

— Paul Halmos

“What I cannot create, I do not understand.”

— Richard Feynman

This project is **not a competition**. You are evaluated on your own growth, not relative to others.

Philosophy (cont.)

- Choose a topic that **genuinely interests you**.
- Calibrate difficulty to **your** proficiency in math and coding.
- A polished small project beats an ambitious incomplete one.

Overview

- **Timeline:** Deadlines and milestones
- **Required Elements:** Learning, Analysis, and Research
- **The Proposal:** Setting a concrete foundation
- **Policies:** AI usage and peer evaluations

Timeline

Milestone	Date	Deliverable
Project Proposal	April 22	1–3 page document
Final Presentations	May 11	In-class demo / slides
Final Submission	May 11	Code, report, & slides

Required Project Elements

To receive full credit, your project must balance three elements:

1. **Learning:** Expanding your toolkit
2. **Analysis:** Testing the boundaries
3. **Research:** Creative modification

Learning

- **Goal:** Bridge the gap between class and the real world.
- Apply class concepts to a novel domain (e.g., finance, biology, physics).
- **OR** Learn a new algorithm or method not covered in the syllabus.
- *Requirement:* Identify and study at least 2–3 foundational references.

Analysis

- **Definition:** Clearly define your hypothesis, assumptions, and context.
- **Transparency:** Analyze the model / algorithm and its key parameters.
- **Methodology:** This can be a theoretical derivation or a rigorous empirical study.

Research

- **The “Delta”:** Propose a modification to the model or algorithm.
- **Evaluation:** Rigorously analyze the modification.
- **Reflection:** Evaluate whether outcomes improved.
- *Note:* A “negative” result is scientifically valid if the reasoning is sound.

Example Projects

Example: Fast Inverse Square Root (Quake III)

- **Learning:** Study the `0x5f3759df` bit-manipulation trick. Understand IEEE 754 floats and why the “magic number” works.
- **Analysis:** Compare speed vs. accuracy against `numpy.sqrt`. Analyze error across input ranges and Newton-Raphson iterations.
- **Research:** Can you find a better magic constant? Adapt the technique for $x^{-1/3}$? Evaluate tradeoffs.

Example: Stock Volatility with Monte Carlo

- **Learning:** Apply variance reduction techniques (antithetic variates) to a finance problem. Study 2–3 papers on Monte Carlo in finance.
- **Analysis:** Compare convergence rates with/without variance reduction across 3 parameter settings. Justify parameter choices.
- **Research:** Propose a hybrid sampling strategy combining antithetic variates with control variates; test whether it improves efficiency.

Example: Infectious Disease Spread (SIR Model)

- **Learning:** Study the SIR compartmental model. Understand the role of transmission rate β and recovery rate γ in disease dynamics.
- **Analysis:** Implement and compare solvers (Euler, RK4). Test sensitivity to parameters and initial conditions. Validate against known analytical behavior.
- **Research:** Extend to SEIR (add “exposed” compartment) or add vaccination. Does your modification change long-term dynamics?

The Proposal

Due: April 22 (1–2 pages)

Your proposal must include:

- **Project Scope:** Explain *your understanding* of the project.
- **Contribution Matrix:** Concrete, actionable items assigned to each group member.
- **Logistics:** Proposed meeting dates and times.
- **References:** Initial bibliography (2–3 items).

Contribution Matrix

Your matrix should specify **who owns what**.

 **Good:**

Member	Responsibility
Alice	Code implementation of base algorithm
Bob	Write-up of Sections 1–2; literature review
Charlie	Design and run experiments; generate figures

 **Not Helpful:**

“We’ll all work on everything together and split tasks as needed.”

Presentation Guidelines

Format

- **Length:** 10–12 minutes + 3 minutes Q&A
- **Required Sections:**
 1. Problem setup and motivation
 2. Methods (what you learned and applied)
 3. Results (analysis and research components)
 4. Reflection (what worked, what didn't, what you learned)

Logistics

- **Live demo** optional but encouraged if feasible.
- **All group members** should speak.
- Slides will be submitted with final materials.

Collaboration Tools

Purpose	Tool	Notes
Document Sharing	OneDrive / Google Drive	Shared folders for notes, data, references
Writing	Overleaf	Real-time collaborative LaTeX; free with .edu email
Code	Google Colab	Shared Jupyter notebooks; no setup required
Code	Git / GitHub	Version control; preferred

AI Usage Policy

AI tools are **allowed** but use them wisely.

-  Use AI to **learn concepts**, debug code, brainstorm ideas.
-  Use AI to **improve writing** clarity and grammar.
-  Do not use AI to **generate work you don't understand**.

 **Warning:** If we suspect blind AI usage without understanding, you will receive an **Incomplete** until you pass a one-on-one oral exam about your project.

Peer Evaluations

To ensure fair contribution, you will **grade your group members** at the end of the project.

Aspect	What You'll Evaluate
Participation	Did they attend meetings and communicate?
Contribution	Did they complete their assigned tasks?
Collaboration	Were they responsive and supportive?

 Evaluations are **private**. I will not share individual responses with your group.

Questions?

Office hours: Tuesday 11:30 am – 4:30 pm in Wyman N452

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